2026-2학기 세포의 분자적 이해

10/1 (목)

Chapter 9.
Protein Synthesis, Processing, and Regulation (3)

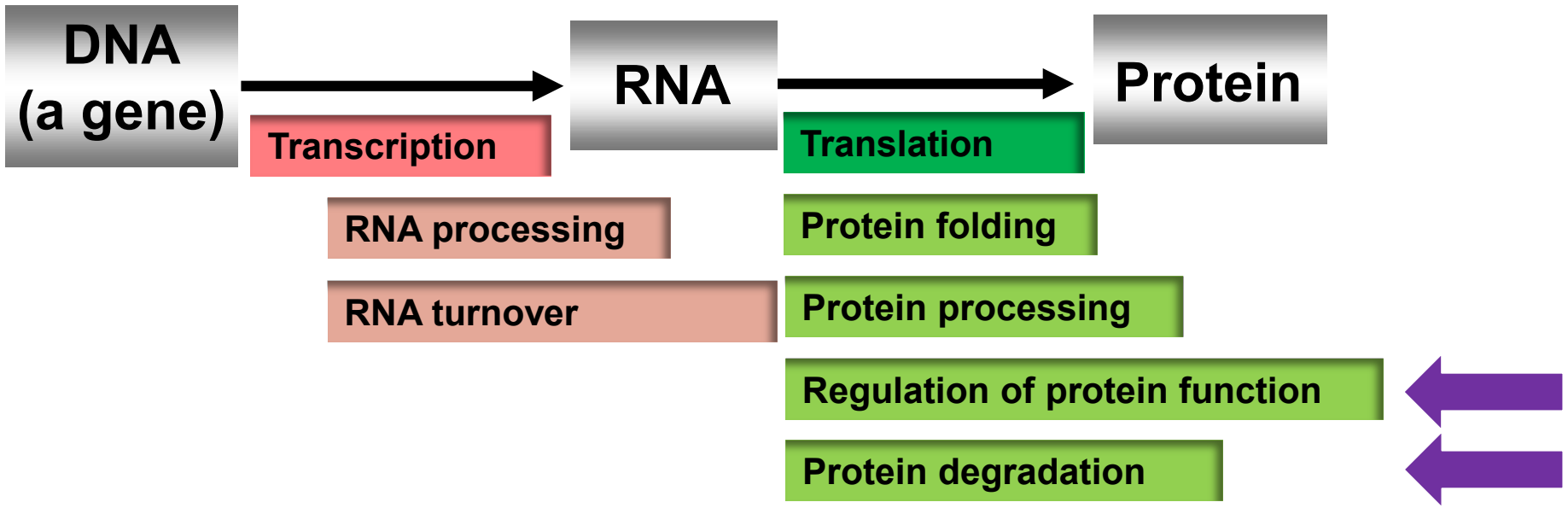
분자의학교실 장병철

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Lecture Contents in part 3

4. Regulation of protein function in eukaryotes

5. Protein degradation in eukaryotes



4. Regulation of protein function

▶ Regulation of protein function allows the cell to regulate not only the amounts but also the activities of its cellular proteins.

▶ There are three general mechanisms of control of cellular proteins:

- Regulation of protein function by small molecules (eg, cAMP)
- Regulation of protein function by phosphorylation
- Regulation of protein function by protein-protein interactions

4. Regulation of protein function

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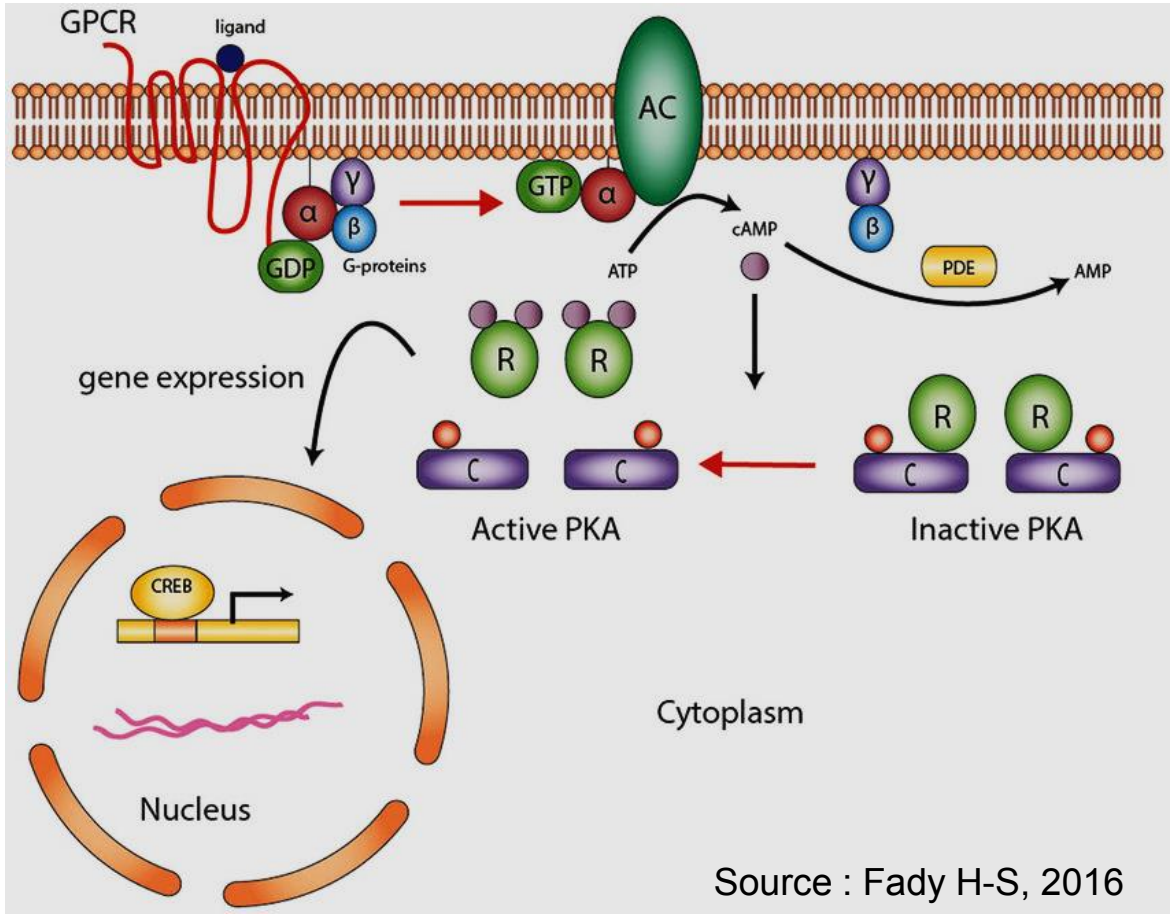
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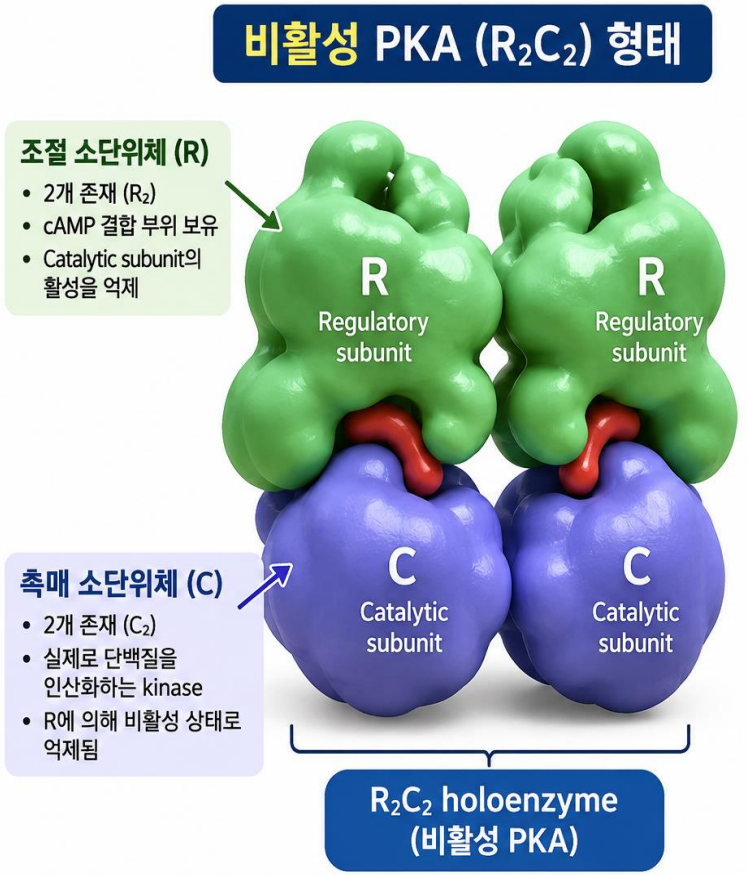
4. Regulation of protein function in eukaryotes

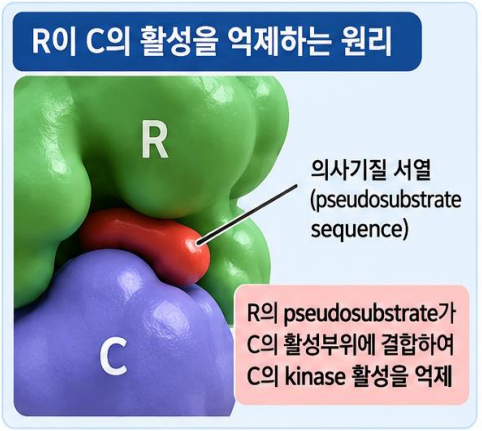
4.1. Regulation of protein function by small molecules

▶ **Most enzymes** [protein kinase A (**PKA**), etc] are controlled by changes in conformation, often as a result of binding small molecules such as **cyclic adenosine monophosphate (cAMP)**, when cells are stimulated by ligands.



Source : Fady H-S, 2016





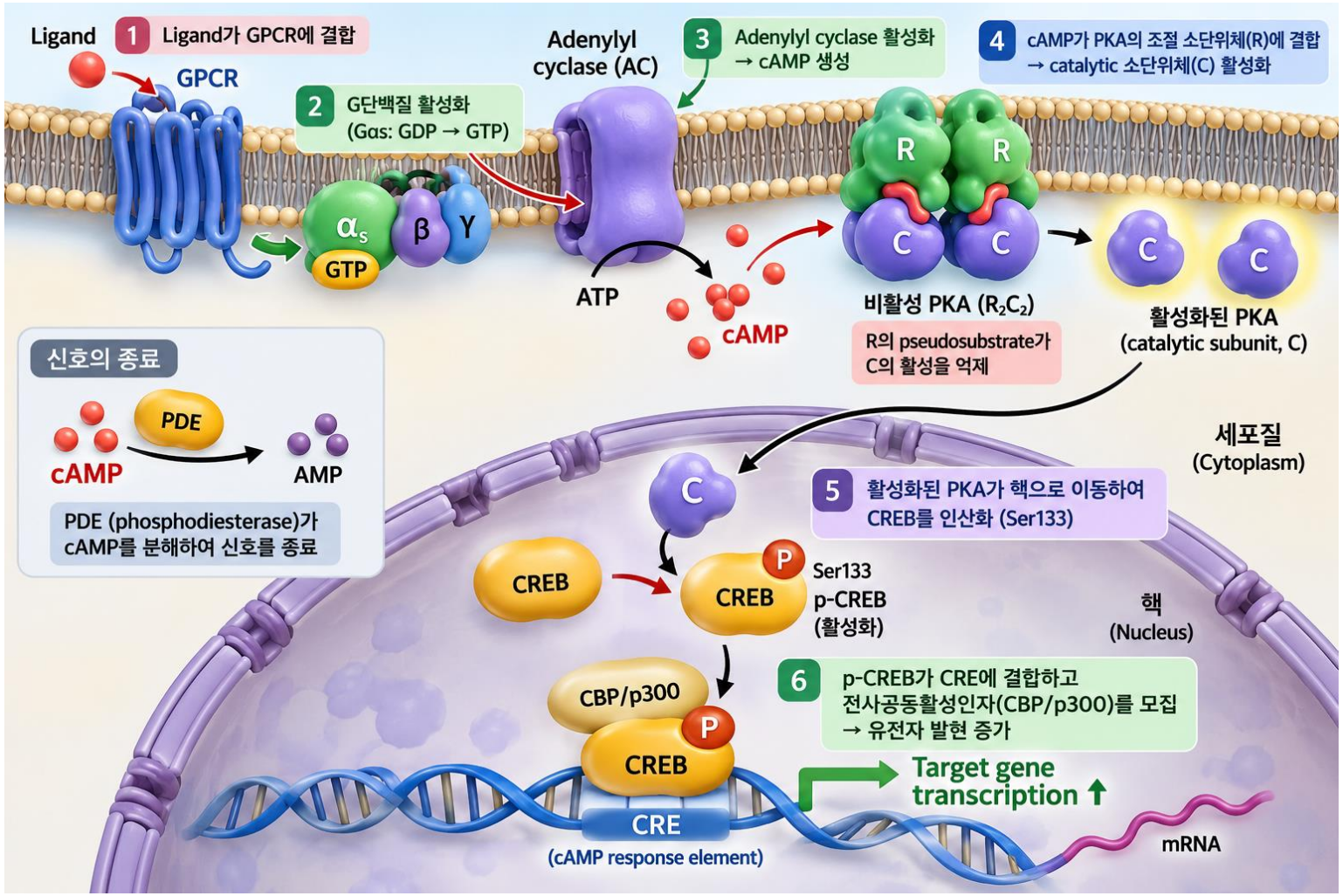
비활성 PKA (R_2C_2)의 특징

구성	Regulatory subunit 2개 (R_2) + Catalytic subunit 2개 (C_2)
상태	비활성 상태 (inactive)
억제 기전	R의 pseudosubstrate가 C의 활성부위에 결합하여 억제
cAMP 결합 부위	R subunit에 존재 (각 R에 2개의 결합 부위)
기능	세포 내 cAMP 농도가 낮을 때 PKA 활성을 억제하여 신호를 차단

4. Regulation of protein function in eukaryotes

4.1. Regulation of protein function by small molecules

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4. Regulation of protein function

▶ Regulation of protein function allows the cell to regulate not only the amounts but also the activities of its cellular proteins.

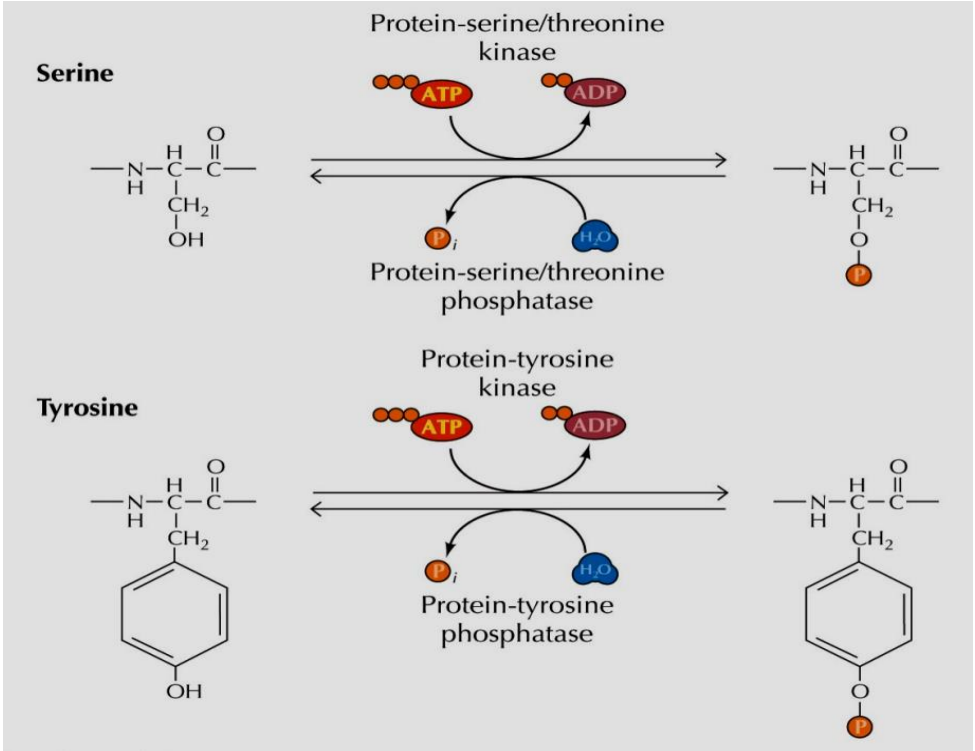
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4. Regulation of protein function in eukaryotes

4.2. Regulation of protein function by phosphorylation

- **Protein phosphorylation** is **a reversible covalent modification process that can activate or inhibit a wide variety of proteins in response to environmental signals/stresses**.
- **Protein kinases** are the enzymes that transfer phosphate groups from ATP to the hydroxyl (-OH) groups of side chains of serine (Ser, S), threonine (Thr, T), or tyrosine (Tyr, Y).
- Protein phosphorylation is reversed by **protein phosphatases**, which catalyze hydrolysis of phosphorylated amino acids.

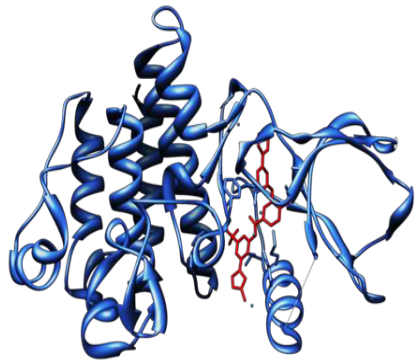


4. Regulation of protein function in eukaryotes

● Protein kinases (a target enzyme for new drug development)

- ▶ Protein kinases are often **components of signal transduction pathways**.
- ▶ **The sequential action of a series of protein kinases** can transmit a signal received at the cell surface to target proteins within the cell, resulting in changes in cell behavior in response to environmental stimuli.
- ▶ **Aberrations in signaling pathways**, especially involving protein tyrosine kinases are responsible for **some kinds of cancer**.
- ▶ **The first protein-tyrosine kinase was discovered in 1980** in studies of Rous sarcoma virus.
- ▶ **Small molecule inhibitors of these enzymes** are promising drugs for cancer treatment (eg, **a Bcr-Abl kinase inhibitor**; **Gleevec**).







4. Regulation of protein function

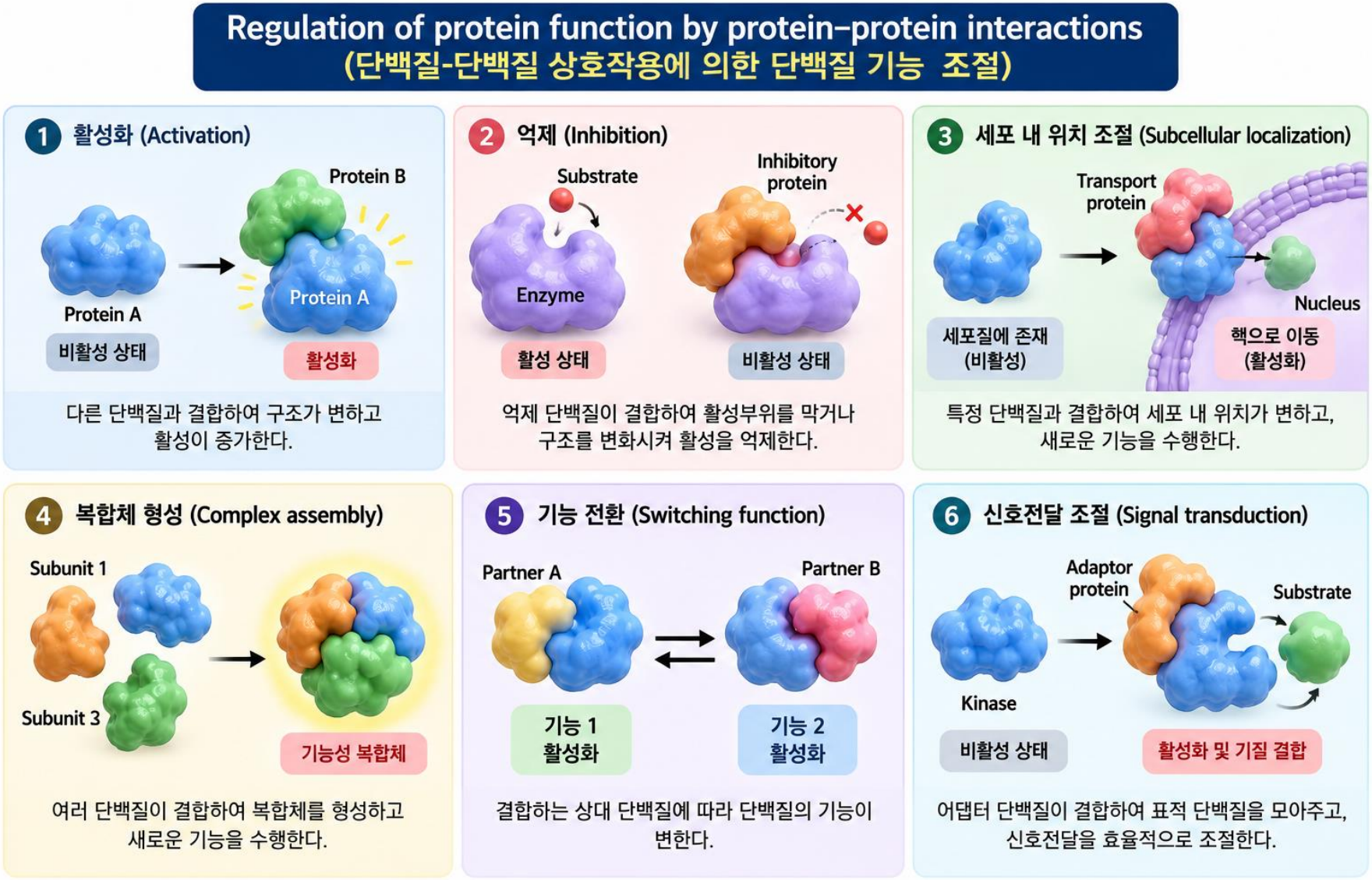
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4. Regulation of protein function in eukaryotes

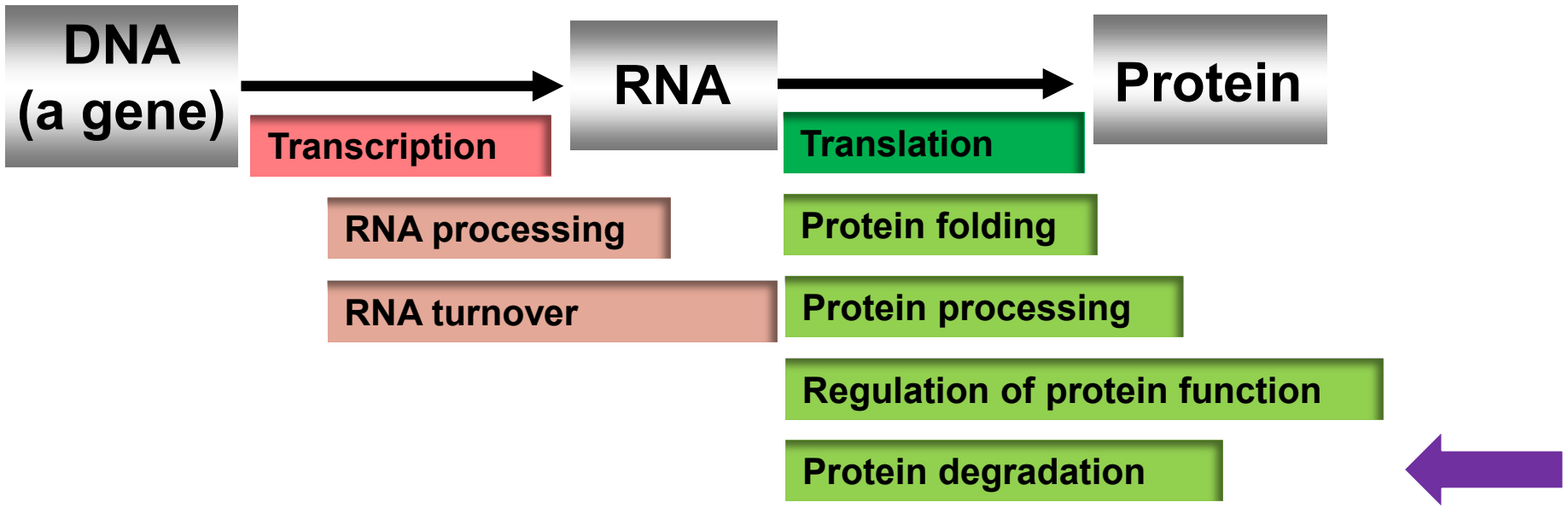
4.2. Regulation of protein function by protein-protein interactions



Lecture Contents in part 3

4. Regulation of protein function in eukaryotes

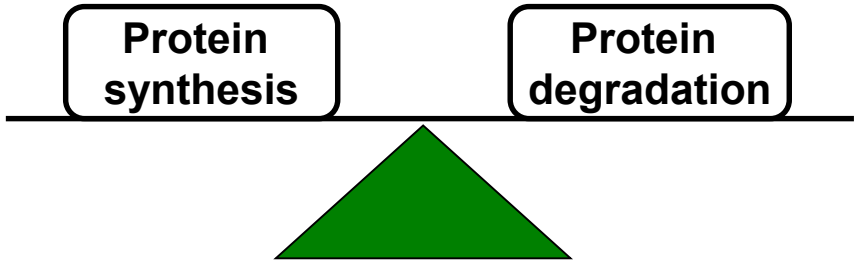
5. Protein degradation in eukaryotes



5. Protein degradation in eukaryotes

5.1. The concept of protein degradation

(1) Proteins levels in cells are determined by rates of synthesis and rates of degradation.



(2) Differential rates of protein degradation are an important aspect of cell regulation.

(3) Many regulatory proteins have short half lives; this allows levels to change quickly in response to external stimuli.

(4) Faulty (nonfunctional) or damaged proteins are recognized and rapidly degraded.

5.2. Major protein degradation pathways

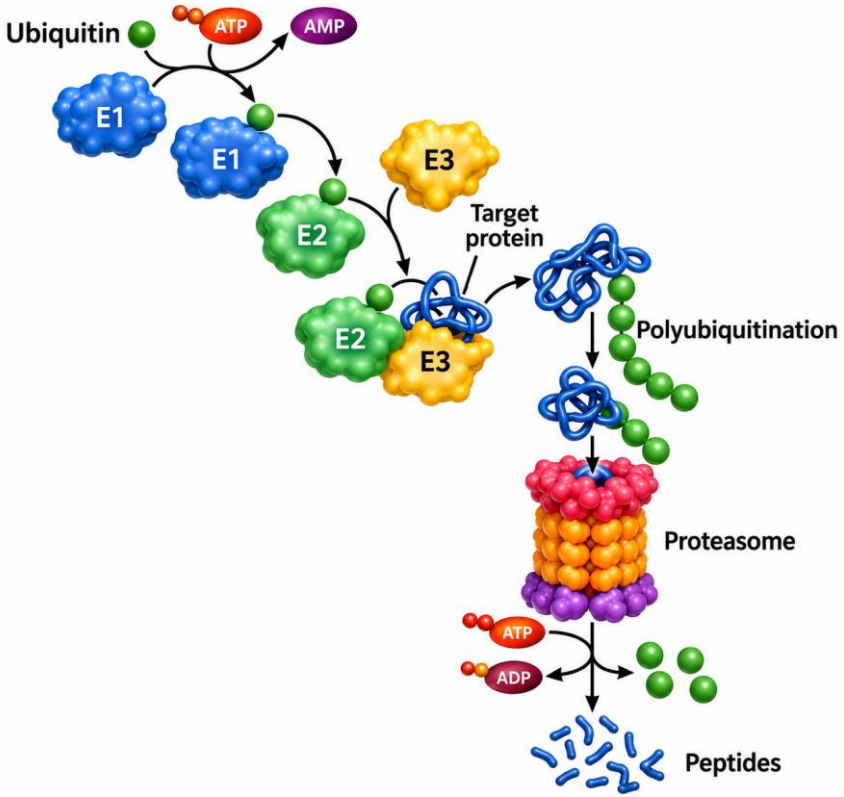
(1) Ubiquitin-proteasome pathway - Utilizes ubiquitin and proteasome

(2) Lysosomal pathway - Use lysosome

5. Protein degradation in eukaryotes

(1) Ubiquitin-proteasome pathway

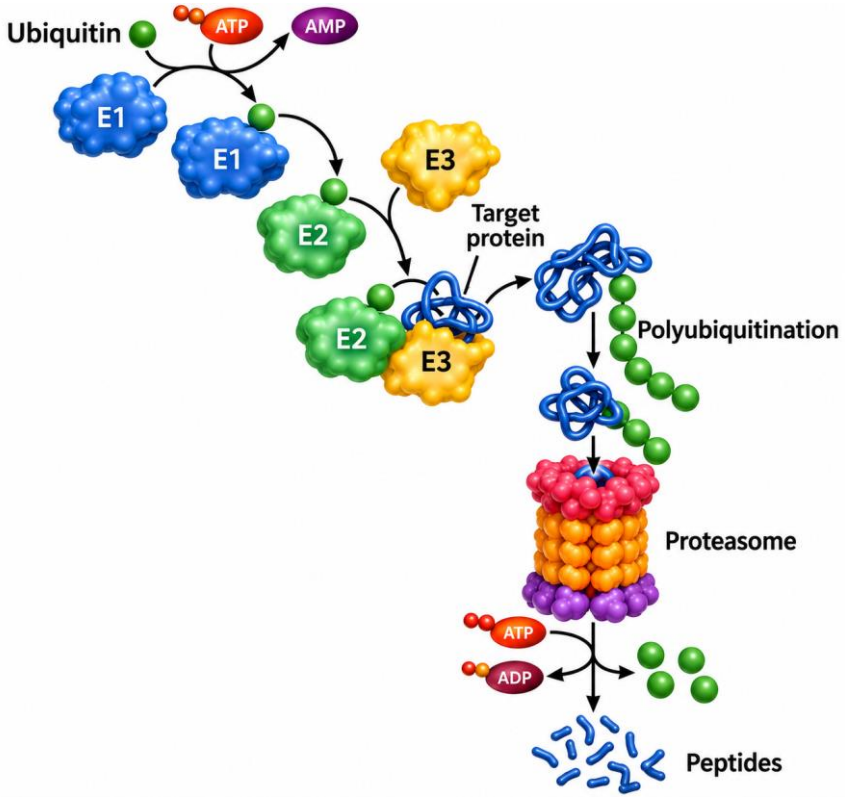
- **Ubiquitin (Ub) is a small protein (polypeptide), consisting of 76 amino acids**, that is found in all cellular tissues in humans, which **helps to regulate the processes of proteins in the body**.
- **Ub is a marker that targets proteins for rapid proteolysis**.
- Ub is **attached to the amino group of the side chain of a lysine residue**, then more are **added to form a chain**.
- Resultantly, **polyubiquinated proteins** are recognized and degraded by a large protease complex, **the 26S proteasome**.

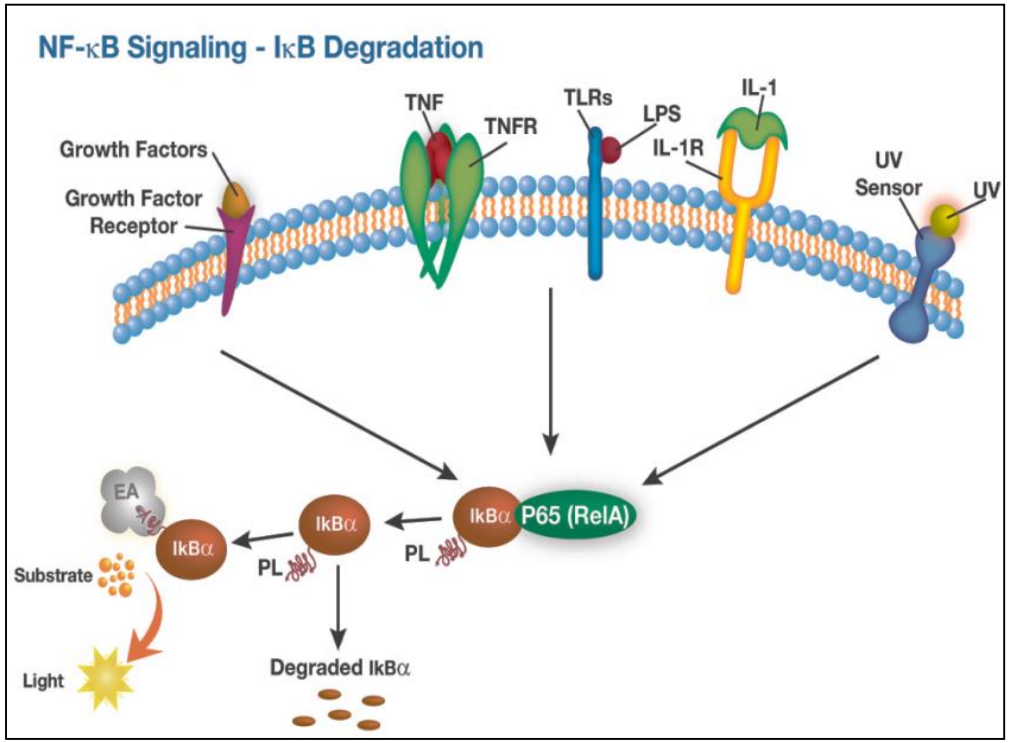


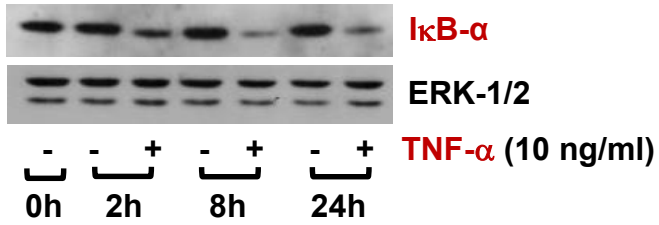
- **Ubiquitination** is a multistep process involving several enzymes : E1, E2, and E3.
- The specificity of E3 enzymes selectively targets proteins for degradation.
- **E1 (activating enzyme)**: Ubiquitin reacts with ATP to form ubiquitin-AMP, and then a thioester bond is created between ubiquitin and E1.
- **E2 (conjugating enzyme)**: E2 receives ubiquitin from E1 and carries it to E3.
- **E3 (ligase)**: E3 specifically recognizes the target substrate protein and attaches ubiquitin to it.

5. Protein degradation in eukaryotes

(1) Ubiquitin-(26S) proteasome pathway (continued)



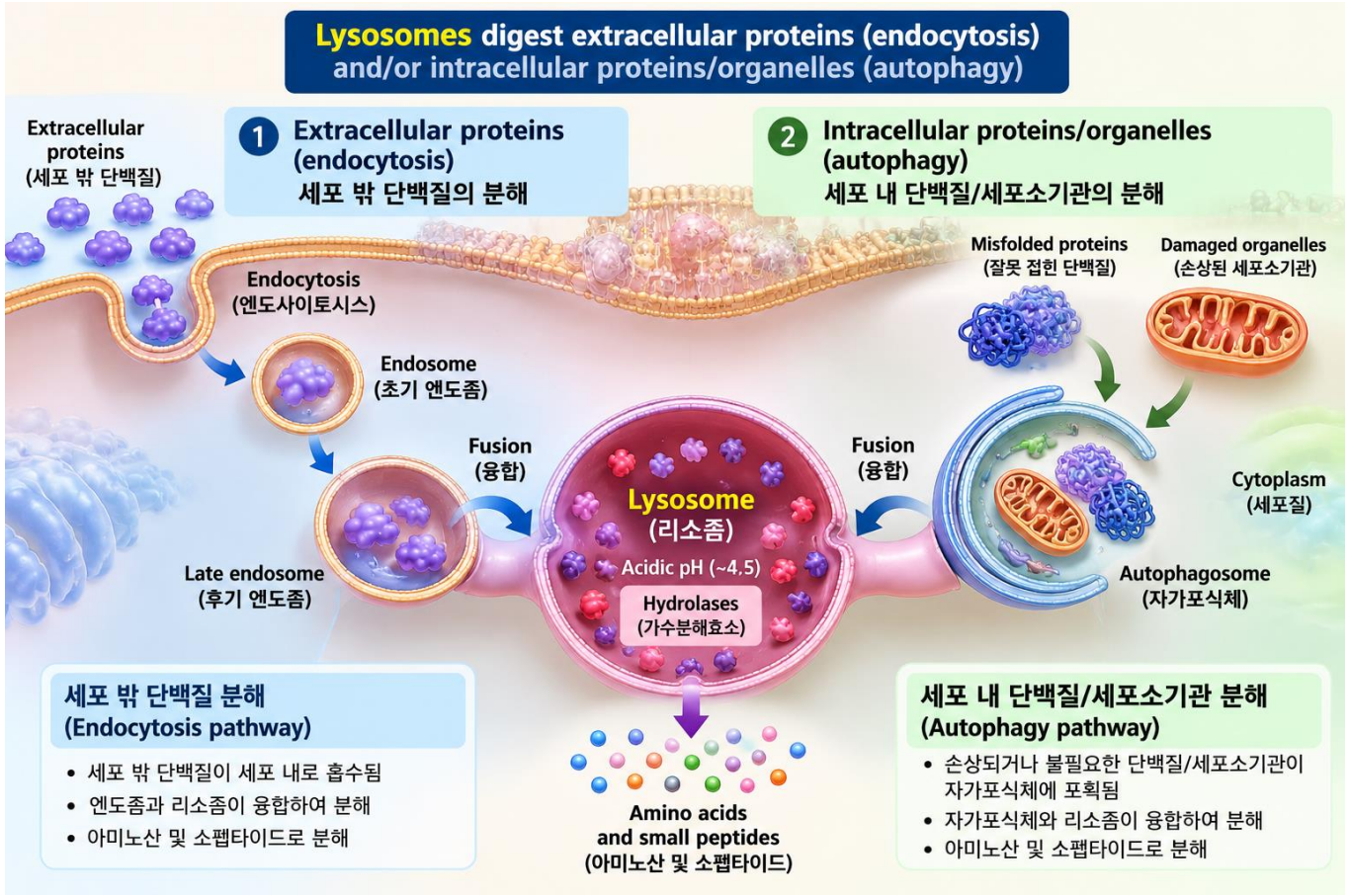


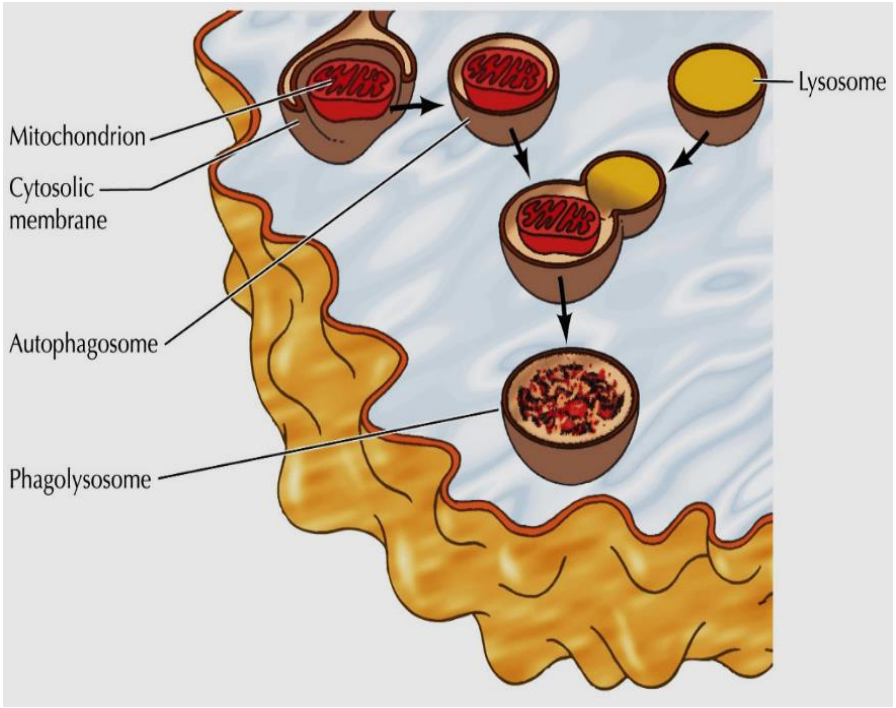


5. Protein degradation in eukaryotes

(2) Lysosomal pathway

- **Protein degradation** can also take place in **lysosomes** - membrane-enclosed organelles that **contain digestive enzymes, including proteases**.
- **Lysosomes** digest **extracellular proteins** taken up by endocytosis, and/or **intracellular proteins/organelles**; and **take part in turnover of these proteins and organelles**.

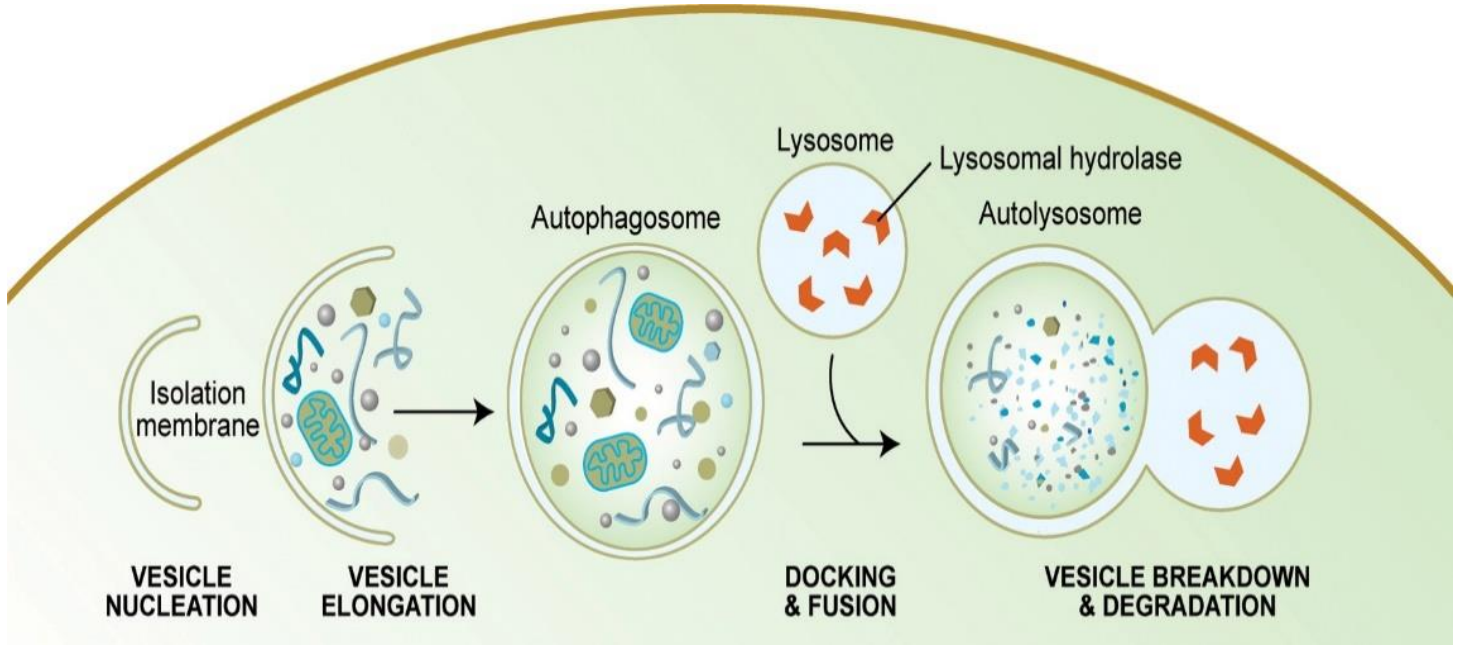




5. Protein degradation in eukaryotes

■ Autophagy

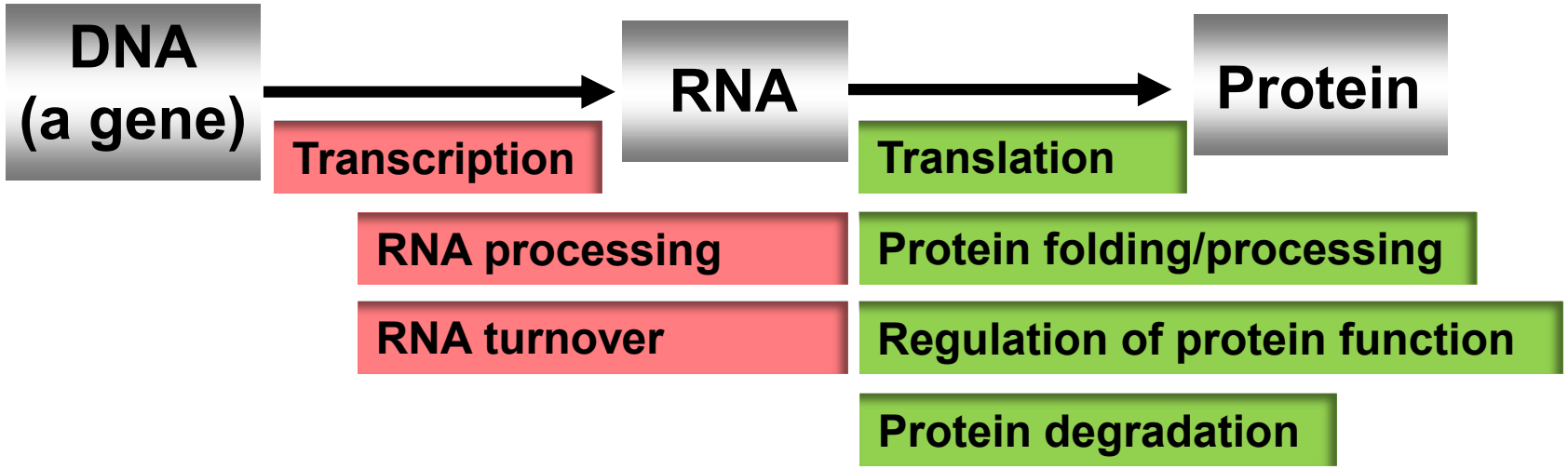
- **A catabolic process involving the degradation of a cell's own components** through the lysosomal machinery
- Can be induced in cells exposed to **nutrient starvation (serum withdrawal)**, allowing cells to degrade nonessential proteins and organelles and reutilize the components.
- **Autophagy plays a role in many cellular processes, including development, cell growth, and apoptosis**
- **In cancer biology, autophagy may be used as an alternative treatment in destroying (or killing) certain cancer cells that have resistance to anti-cancer drug** (→ "autophagic cell death" = type II programmed cell death).



Lecture Contents

1. Introduction about learning performance of today's lecture
2. Protein folding in eukaryotes
3. Protein processing in eukaryotes
4. Regulation of protein function in eukaryotes
5. Protein degradation in eukaryotes

Chapter 8/9. RNA / Protein Synthesis, Processing, and Regulation



감사합니다!